

IOT APPLICATIONS DESIGN

Course Code: CSI 601

Credit Units: 03
Total Hours: 30

Course Objectives:

- Ability to understand the coding methodology for IOT.
- Understand the software framework for IoT
- Ability to understand the hardware to hardware communication and hardware to software communication
- Understand the various types of advance input/output interface.

Course Contents:

Module I: Introduction: (8 Hours)

Introduction – History - Creative Coding Platforms - Open Source Platforms – PIC - Arduino, Sketch, Raspberry Pi, Iterative coding methodology – Python Programming - Mobile phones and similar devices - Arm Devices - Getting used to Arduino - Sensor Characterization: Safety, Basic Electronics (circuit theory, measurements, parts identification) Sensors and Software: Understanding Processing Code Structure, variables and flow control, Interfacing to the Real World

Module II: Software Frameworks: (6 Hours)

Software: open Frameworks as our IDEs - “Arduino” Languages - Hardware: Desktop / Laptop / Raspberry Pi - How to approach a programming problem? Representing “reality” with computers. Digital vs. Analog circuits, audio, communication, etc. Analog to Digital Conversion (ADC) - Digital to Analog Conversion (DAC) – Microcontrollers, Python Programming for IoT.

Module III: Hardware Communication: (8 Hours)

Communication – Serial& Parallel - Hardware to Hardware Communication - I2C/IIC (Inter-Integrated Circuit) - SPI (Serial Peripheral Interface) – Serial UART Communication - Introduction to the command line – git/github. Introduction to Programming: A comparative studio between Arduino and open Frameworks - Arduino-compatible Microcontrollers Sensors and Actuators

Module IV: Advanced I/O Interfacing: (8 Hours)

Advanced I/O – open Frameworks: Live Network feeds (push and pull) - Data persistence (saving data and preferences) - Database interface (MySQL, sqLite, XML, PHP/Web) - Arduino: Wired/Wireless Networking

UNIT V - IoT, FUTURE AND PERSPECTIVES (6 hours) Talking to the cloud: Baby steps to Internet of Things, TCP/IP and UDP - Building peer to peer communication system using Bluetooth & WiFi.

Course Outcomes:

The student will learn

- Understand the concept of IOT Open Source Platform.
- Understand software platform for IOT.
- How use & Access hardware communication for IOT.
- How can send and stored data in database which is sense by sensor from remote location.

Examination Scheme:

Components	A	CT	S/V/Q/HA	EE
Weightage (%)	5	15	10	70

A: Attendance, CT: Class Test, S/V/Q/HA: Seminar/Viva/Quiz/ Home Assignment, EE: End Semester Examination;

Text & References:

- Internet of things programming projects by colin dow: Build IoT solution with Raspberry Pi 3 and Python
- Hands on Internet of Things with Blynk by Pradeeka Seneviratne
- Industrial Internet Application Development by Alena Traukina, Jayant Thomas

IOT APPLICATIONS DESIGN LAB

Course Code: CSI 621

Credit Unit: 01

Total Hours: 20

Course Objectives:

- Ability to understand the coding methodology for IOT.
- Understand the software framework for IoT
- Ability to understand the hardware to hardware communication and hardware to software communication
- Understand the various types of advance input/output interface.

Programs List:

1. Write a program for Arduino board which takes input from infra red sensor and display count on 16 X 2 LED screen: **(2 Hours)**
2. Write a program to send SMS on a mobile phone from Arduino board using GSM module: **(2 Hours)**
3. Write a program to receive call from any mobile using Android board and GSM module: **(2 Hours)**
4. Write a program to sense temperature at remote location using Arduino board and temperature sensor: **(2 Hours)**
5. Write a program for Arduino board to sense moisture at some remote location using moisture sensor: **(2 Hours)**
6. Write a program for Arduino board to create smart burglar alarm using suitable sensors: **(2 Hours)**
7. Design a circuit board for water level detector system using suitable microcontroller and sensors: **(2 Hours)**
8. Design and program a complete smart home system using appropriate sensors, actuators and microcontroller unit for automatic regulation of home electrical appliances like room bulbs, ceiling fans etc: **(2 Hours)**
9. Consider a customer who wants automatic water pump system for his home such that the system must be able to switch on and off water pump automatically without his intervention. Design and program the complete system using appropriate sensors, actuators and microcontroller unit: **(2 Hours)**
10. Consider customer want a dedicated small power web server for his home automation system such that he/she can communicate between his/her various home appliances. Design and program a complete smart home dedicated web server system using appropriate sensors, actuators and microcontroller unit: **(2 Hours)**

Course Outcomes:

The student will learn

- Understand the concept of IOT Open Source Platform.
- Understand software platform for IOT.
- How use & Access hardware communication for IOT.
- How can send and stored data in database which is sense by sensor from remote location.

Examination Scheme:

IA			EE			
A	PR	Practical Based Test	Major Experiment	Minor Experiment	LR	Viva
5	10	15	35	15	10	10

Note: IA –Internal Assessment, EE- External Exam, A- Attendance PR- Performance, LR – Lab Record, V –Viva.

Text & References:

- Internet of things programming projects by colin dow: Build IoT solution with Raspberry Pi 3 and Python.
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